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Department of CSE

B.Tech. CSE Trimester -5

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Generative AI and AI Platforms

25CS2207E

WEEK 1 – FIRST LLM API CALL USING OPENAI

1. Aim

To establish a connection between a Python application and the OpenAI platform by configuring the API key and installing the required Python libraries.

2. Requirements

- Python
- PyCharm IDE
- OpenAI API key
- OpenAI Python SDK
- python-dotenv library
- Internet connection

3. Theory

OpenAI provides an API that allows developers to integrate Large Language Models (LLMs) into Python applications. A Python program can send text prompts to an AI model and receive AI-generated responses.

An API key is used for authentication while accessing the OpenAI API. The API key should be stored securely and should not be directly exposed in the source code.

The OpenAI Python SDK provides functions to communicate with OpenAI models. The python-dotenv library helps load environment variables stored in a `.env` file.

4. Procedure

1. Created a Python project named Week-1 in PyCharm.
2. Configured a Python virtual environment for the project.
3. Installed the required Python libraries.
4. Created a `.env` file to store the OpenAI API key.
5. Configured the API key using the variable `OPENAI_API_KEY`.
6. Created a Python file named `openai_call.py`.
7. Imported the required libraries and loaded the environment variables.
8. Initialized the OpenAI client using the API key.

9. Wrote a program to send a prompt to the OpenAI model.
10. Executed the program using PyCharm.
11. Observed the API response and identified the billing-related error.

5. Code

File Name: openai_call.py

```
import os
from dotenv import load_dotenv
from openai import OpenAI

# Load environment variables
load_dotenv()

# Get the API key
api_key = os.getenv("OPENAI_API_KEY")

if not api_key:
    raise ValueError("OpenAI API key not found!")

# Initialize the client
client = OpenAI(api_key=api_key)

# Send input to the model
response = client.responses.create(
    model="gpt-5.5",
    input="What is Artificial Intelligence? Explain in simple words."
)

# Print the model response
print("\nLLM Response:")
print(response.output_text)
```

6. Execution

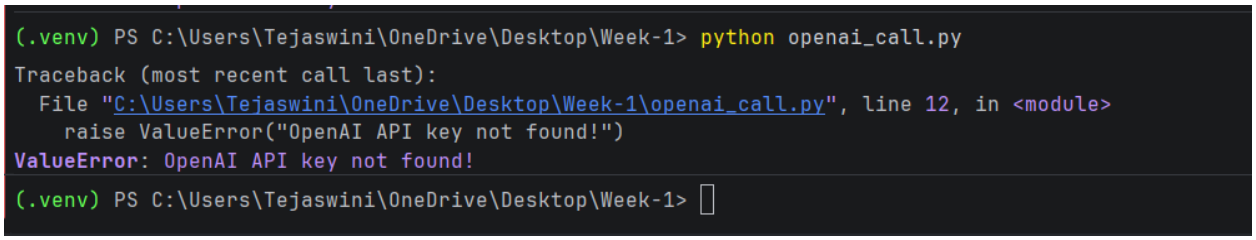
Step 1: Install the required libraries

```
python -m pip install openai python-dotenv
```

Step 2: Execute the Python program

```
python openai_call.py
```

Step 3: Output



7. Observation

- The OpenAI Python SDK was installed successfully.
- The python-dotenv library was installed successfully.
- The API key was configured using a `.env` file.
- The OpenAI client was initialized using the API key.
- The Python program was written to send a prompt to the AI model.
- The API request encountered a billing-related error.
- It was understood that ChatGPT access and OpenAI API access are different services.

8. Result

The initial setup for using the OpenAI API with Python was completed successfully. The required libraries were installed, the API key was configured, and the Python program was prepared and executed. However, the actual LLM response generation could not be completed because the OpenAI API account had no remaining credits.